

Reproducibility in the Life Sciences

Jean-Baptiste Poline

MNI, McConnell Brain Imaging Centre,
McGill, Montreal

Part I: Reproducibility: background

Part II : Etiology of Irreproducibility

Part III :Some therapeutic proposals

Part I: Reproducibility: background

Part II : Etiology of Irreproducibility

Part III :Some therapeutic proposals

- 53 papers examined at Amgen in preclinical cancer research
- Papers were selected that described something completely new and in very high impact factor journals
- **Scientific findings were confirmed in only 6 (11%)**

Begley and Ellis, Nature, 2012



FEATURE ARTICLE



REPRODUCIBILITY IN CANCER BIOLOGY

Challenges for assessing replicability in preclinical cancer biology

- * Not enough information in papers
- * Analytical flexibility is particularly pervasive in neuroimaging analysis (cf workflow tools Nipype, Niflow, Pydra, ...)

Altered Brain Activity in Unipolar Depression Revisited Meta analyses of Neuroimaging Studies

Veronika I. Müller, PhD, Edna C. Cieslik, PhD, Ilinca Serbanescu, MSc, Angela R. Laird, PhD, Peter T. Fox, MD,
and Simon B. Eickhoff, MD

RESULTS—In total, 57 studies with 99 individual neuroimaging experiments comprising in total 1058 patients were included; 34 of them tested cognitive and 65 emotional processing. Overall analyses across cognitive processing experiments ($P > .29$) and across emotional processing experiments ($P > .47$) revealed no significant results. Similarly, no convergence was found in analyses investigating positive (all $P > .15$), negative (all $P > .76$), or memory (all $P > .48$) processes. Analyses that restricted inclusion of confounds (eg, medication, comorbidity, age) did not change the results.

Stein et al., 2012, Nature Genetics, study of the hippocampal volume in more than 10k+7k subjects

Previously identified candidate polymorphisms associated with hippocampal volume in general showed little association within our meta-analysis :(

Stein et al, Nat. Gen. 2013

Essay

Why Most Published Research Findings Are False

John P. A. Ioannidis

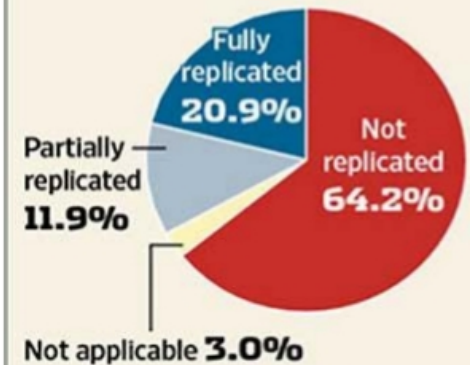
2005. *PLoS Medicine*, 2(8), e124. doi:
10.1371/journal.pmed.0020124

“There is increasing concern about the reliability of biomedical research, with recent articles suggesting that up to 85% of research funding is wasted.”

Bustin, S. A. (2015). The reproducibility of biomedical research: Sleepers awake!
Biomolecular Detection and Quantification

No Cure

When Bayer tried to replicate results of 67 studies published in academic journals, nearly two-thirds failed.



Source: Nature Reviews Drug Discovery



NATURE | NEWS

First results from psychology's largest reproducibility test

THE LANCET

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Research: increasing value, reducing waste

Credibility Crisis

Los Angeles Times | BUSINESS

LOCAL U.S. WORLD BUSINESS SPORTS ENTERTAINMENT HEALTH STYLE TRAVEL

Science has lost its way, at a big cost to humanity

Researchers are rewarded for splashy findings, not for double-checking accuracy. So many scientists looking for cures to diseases have been building on ideas that aren't even true.

Science AAAS.ORG | FEEDBACK | HELP | LIBRARIANS

All Science Journals Enter Search Term

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Science 17 January 2014:
Vol. 343 no. 6168 p. 229
DOI: 10.1126/science.1250475

EDITORIAL

Reproducibility

Marcia McNutt

Marcia McNutt is Editor-in-Chief of Science.

Science advances on a foundation of trusted data. But a new approach that scientists use to gain confidence in their findings was shaken by reports that a troubling number of results are not reproducible. Because confidence in results is essential, we are announcing new initiatives at Science. For preclinical studies (one of the target areas for increasing transparency), authors will indicate handling (such as how to deal with outliers), whether they ensure a sufficient signal-to-noise ratio, whether the experimenter was blind to the conduct of the experiment, and whether they followed the guidelines.

Announcement: Reducing our irreproducibility : Nature News & Comment

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NATURE | EDITORIAL

Announcement: Reducing our irreproducibility

24 April 2013

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Over the past year, *Nature* has published a string of articles that challenge the reliability and reproducibility of published research (collected and

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The meaning of Sachin Tendulkar

HOW SCIENCE GOES WRONG

Ernst Haeckel

TheScientist

EXPLORING LIFE. INSPIRING INNOVATION

NIH Tackles Irreproducibility

The federal agency speaks out about how to improve the quality of scientific research.

By Jef Akst | January 28, 2014

nature International weekly journal of science

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archive > volume 483 > issue 7391 > editorials > article

NATURE | EDITORIAL

Must try harder

Nature 483, 509 (29 March 2012) | doi:10.1038/483509a
Published online 28 March 2012

PDF Citation Reprints Rights & permissions Article metrics

Too many sloppy mistakes are creeping into scientific papers. Lab heads must look more rigorously at the data — and at themselves.

NIH plans to enhance reproducibility

Francis S. Collins and **Lawrence A. Tabak** discuss initiatives that the US National Institutes of Health is exploring to restore the self-correcting nature of preclinical research.

Collins and Tabak. 2014. Nature 505: 612–13.

<https://www.software.ac.uk/>

		Data	
		Same	Different
Code & Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

- My (preferred) way of thinking about reproducibility: **only talk about generalizability** across ... (Data, Software, Time, Scanner, Stimuli, ... etc)

<https://www.software.ac.uk/>

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		Same	Different
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Part III :Some therapeutic proposals

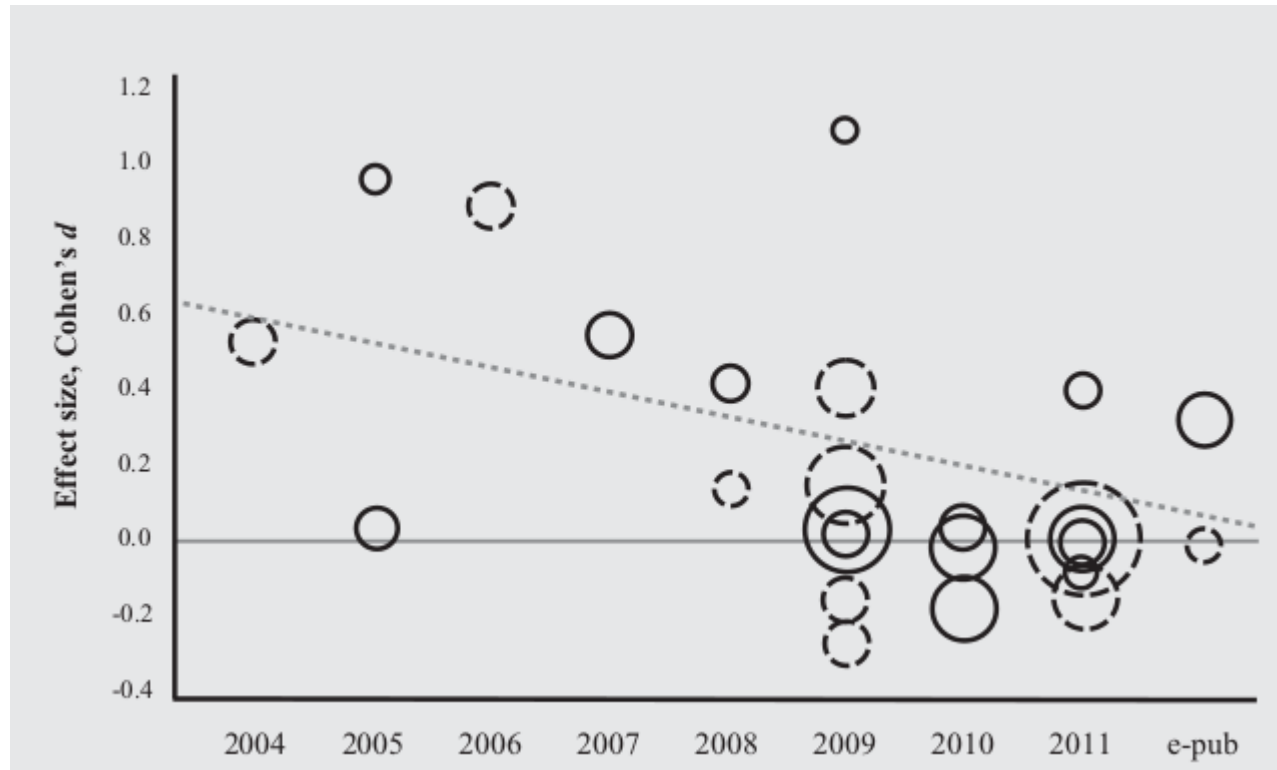
1. Poor statistical procedures
2. Issues in data and software
3. A cultural issue: Publication practices and research incentives

1. Poor statistical procedures

- Small Ns and effect sizes
- Power issues
- P-hacking - harking

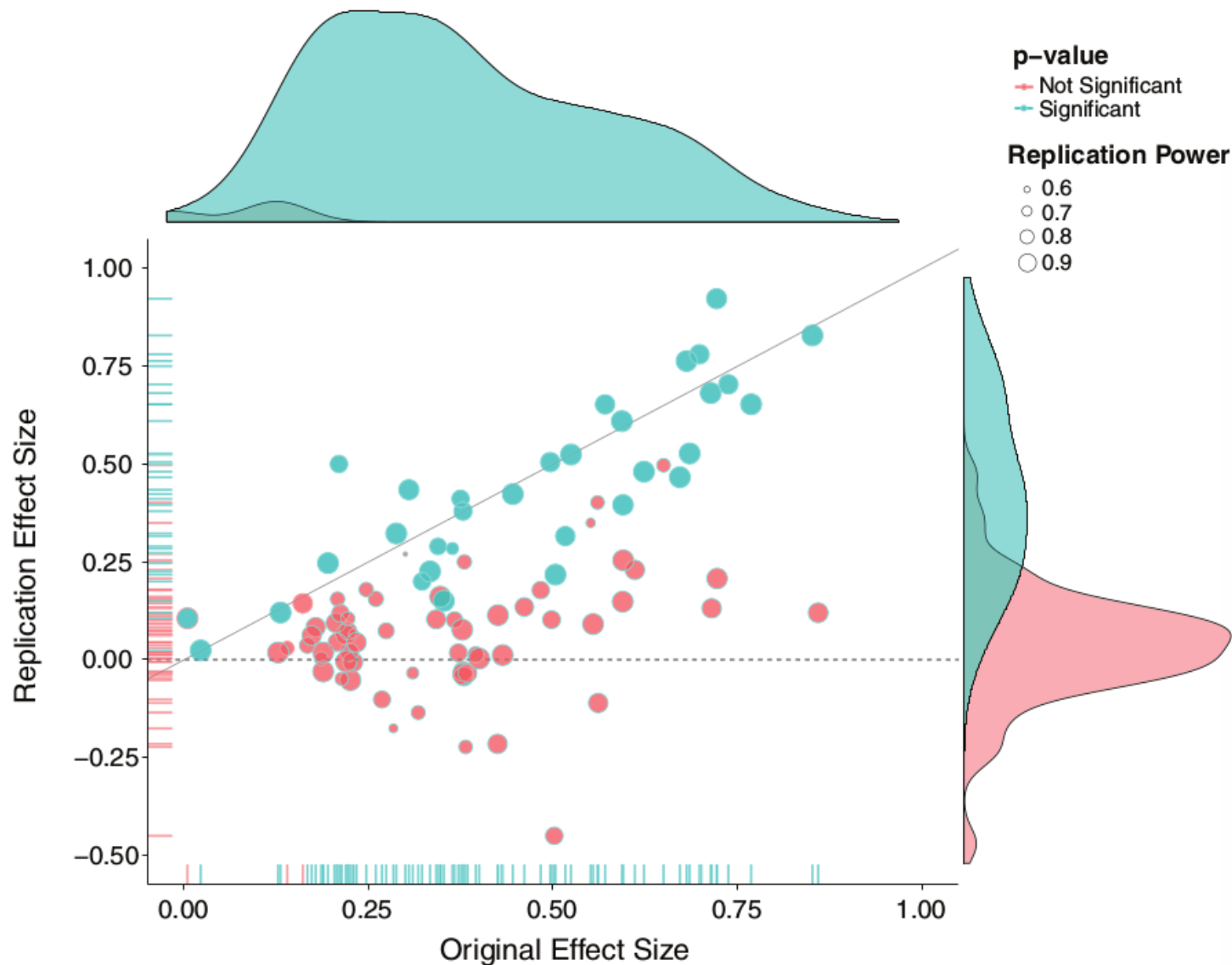
2. Issues in data and software

3. A cultural issue: Publication practices and research incentives



Molendijk, 2012: BDNF and hippocampal volume

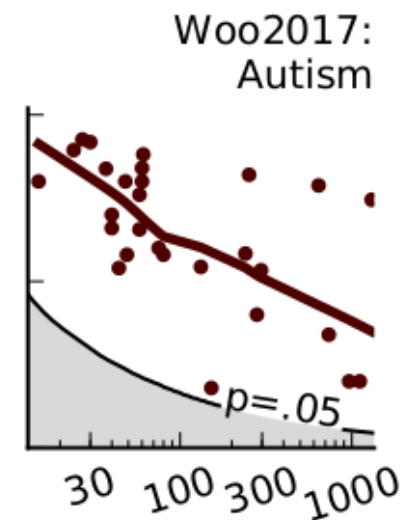
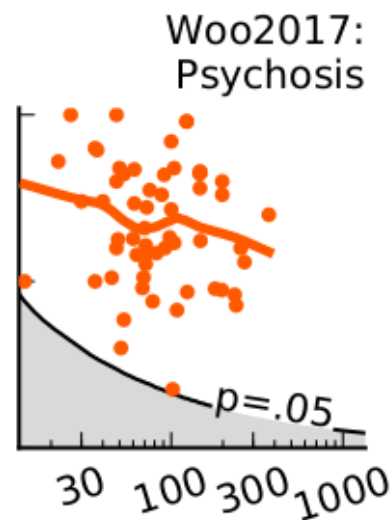
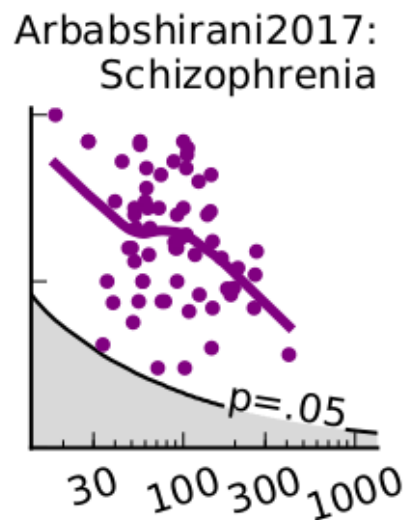
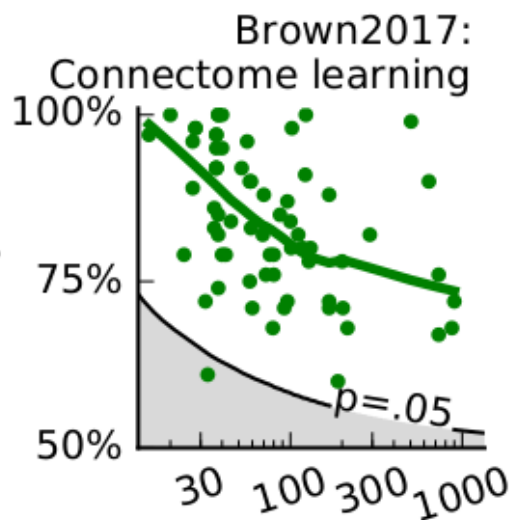
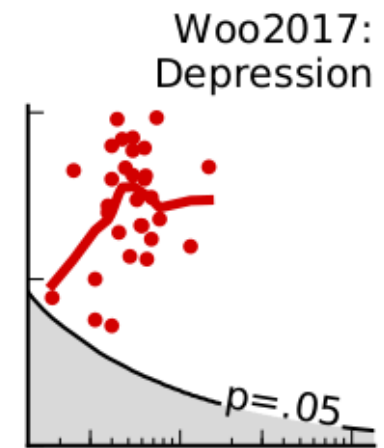
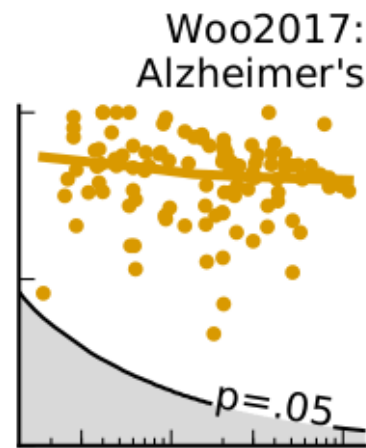
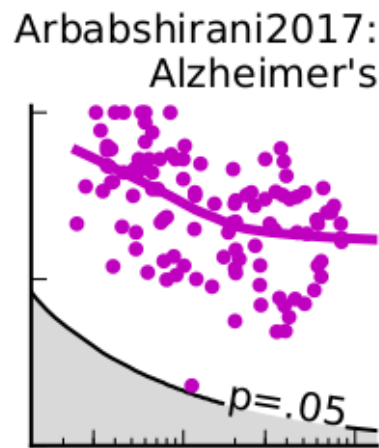
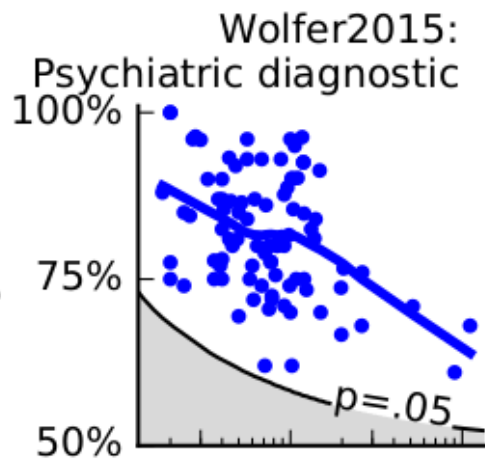
See also : Mier, 2009: COMT and DLPFC



* The mean **effect size** (r) of the replication effects ($M r = 0.197$, $SD = 0.257$) **was half the magnitude** of the mean effect size of the original effects ($M r = 0.403$, $SD = 0.188$)

* **39%** of effects were rated to have replicated the original effect

Reported accuracy



Study sample size

False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant

Joseph P. Simmons¹, Leif D. Nelson², and Uri Simonsohn¹

¹The Wharton School, University of Pennsylvania, and ²Haas School of Business, University of California, Berkeley

Table 1. Likelihood of Obtaining a False-Positive Result

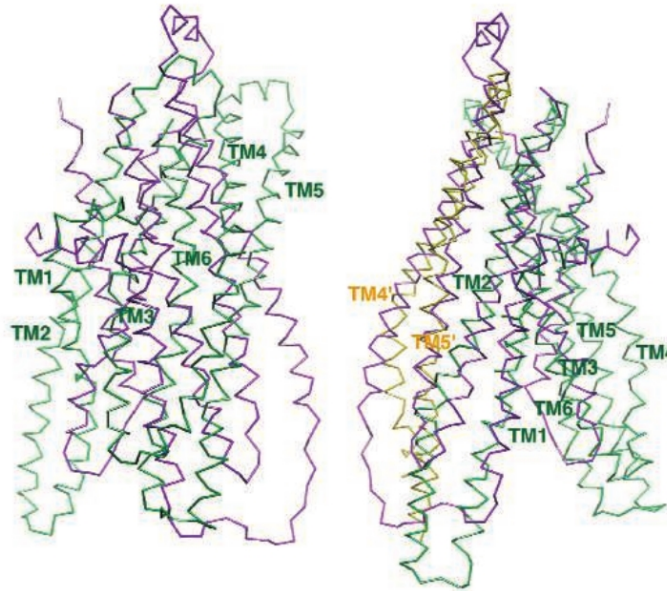
Researcher degrees of freedom	Significance level		
	$p < .1$	$p < .05$	$p < .01$
Situation A: two dependent variables ($r = .50$)	17.8%	9.5%	2.2%
Situation B: addition of 10 more observations per cell	14.5%	7.7%	1.6%
Situation C: controlling for gender or interaction of gender with treatment	21.6%	11.7%	2.7%
Situation D: dropping (or not dropping) one of three conditions	23.2%	12.6%	2.8%
Combine Situations A and B	26.0%	14.4%	3.3%
Combine Situations A, B, and C	50.9%	30.9%	8.4%
Combine Situations A, B, C, and D	81.5%	60.7%	21.5%

1. Poor statistical procedures

2. Issues in data and software

- Research software issues: motivation and examples
- Across OS / implementation
- Across software and parametrization
- Across pipeline: “analytical flexibility”
- Issues in data

3. A cultural issue: Publication practices and research incentives



Flipping fiasco. The structures of MsbA (purple) and Sav1866 (green) overlap little (*left*) until MsbA is inverted (*right*).

- G. Chang: 3 Science, 1 PNAS, 1 J Mol Biol retracted
- “... a homemade data-analysis program had flipped two columns of data...”,
- “... inherited from another lab...”
- The code was distributed and used by others

- Potti et al., Nat. Med. 2006, 2008 vs Baggerly and Coombes, “Forensic analysis”, Annals of applied Stat., 2009
- Choose cell lines that are most sensitive / resistant to a drug, use expression profiles to build a model that predicts patient response

Baggerly and Coombes Forensic:

“with poor documentation and irreproducibility even well meaning investigator may argue for drug that are contraindicated to some patients”

**“the most common errors are simple
(e.g., row or column offsets); conversely,
the most simple errors are common.”**

Across OS

Across implementations

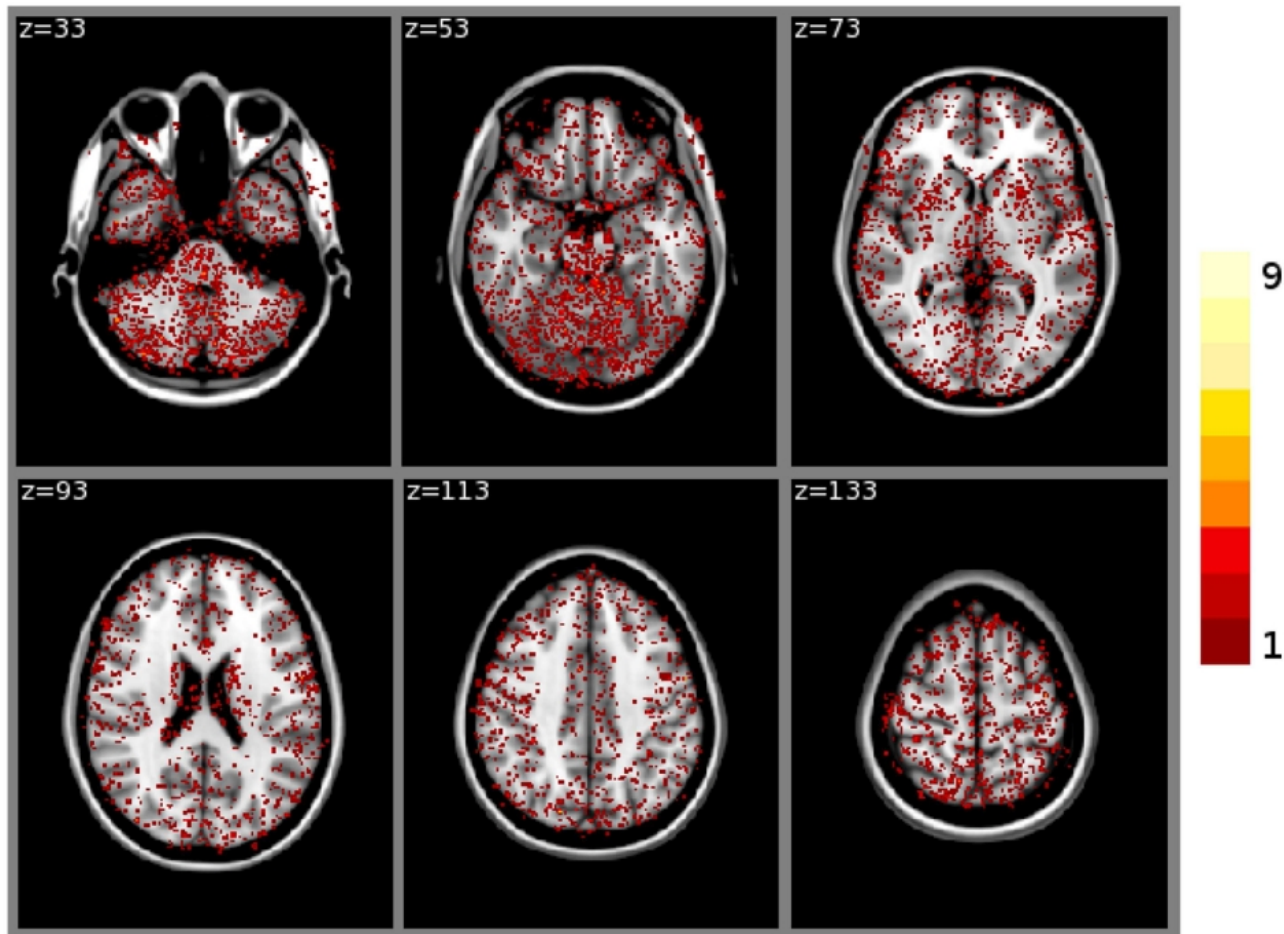
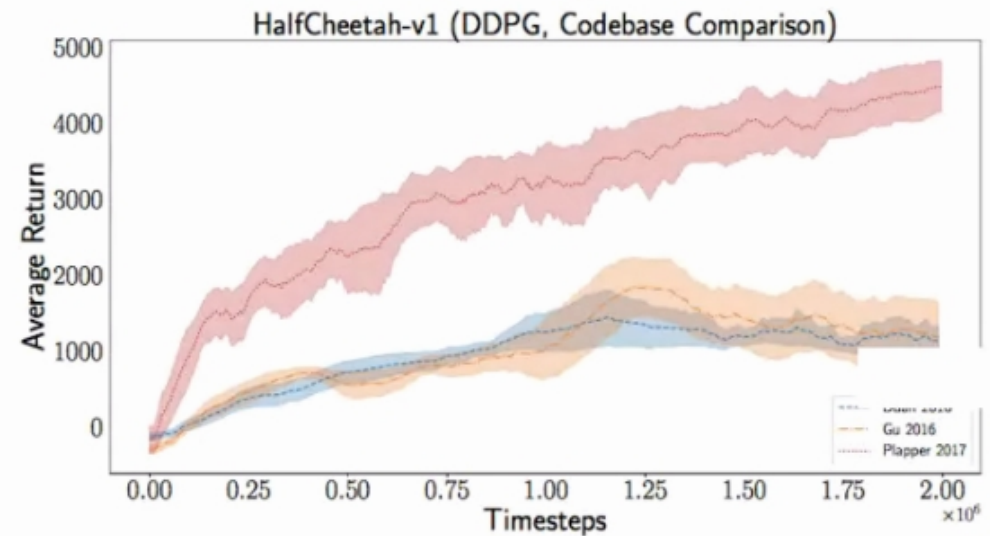
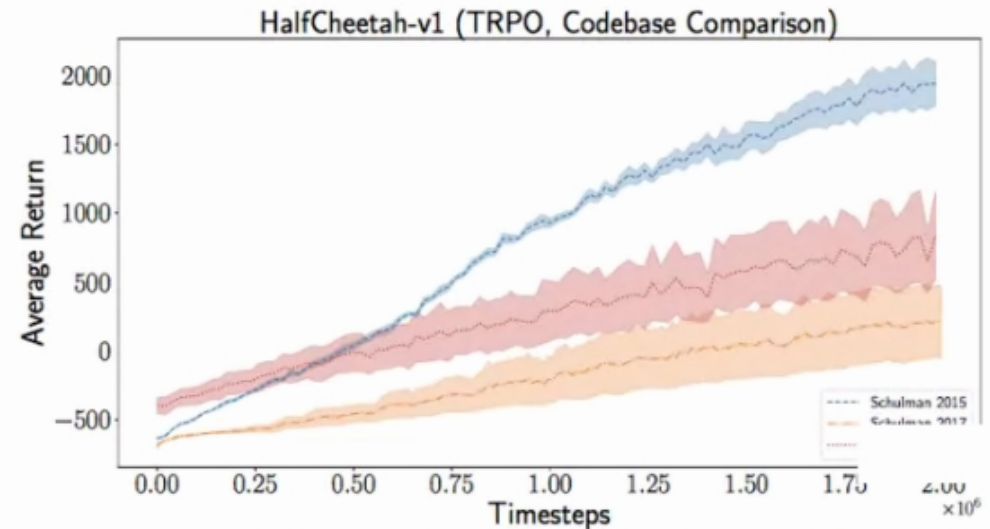


FIGURE 2 | Sum of binarized differences between cortical tissue classifications obtained on cluster A and cluster B (FSL FAST, build 1, $n = 150$ subjects). All binarized differences were resampled to the default MNI152 volume template.

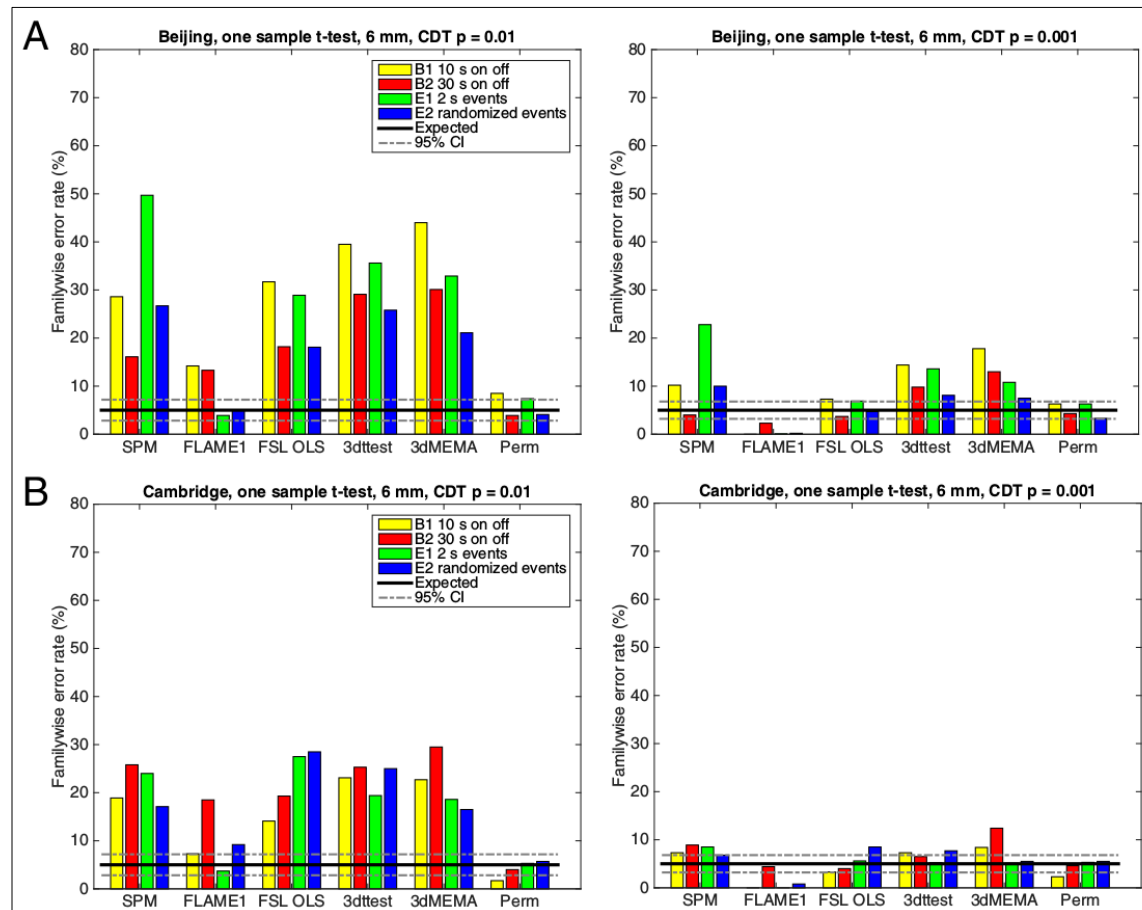
Glatard et al, 2015, F. in Neuroinformatics

- Same algorithm
 - Top graph: TRPO
 - Bottom graph: DDPG
- Same domain
- Simulation environment
- Different implementations



Credit : Joelle Pineau

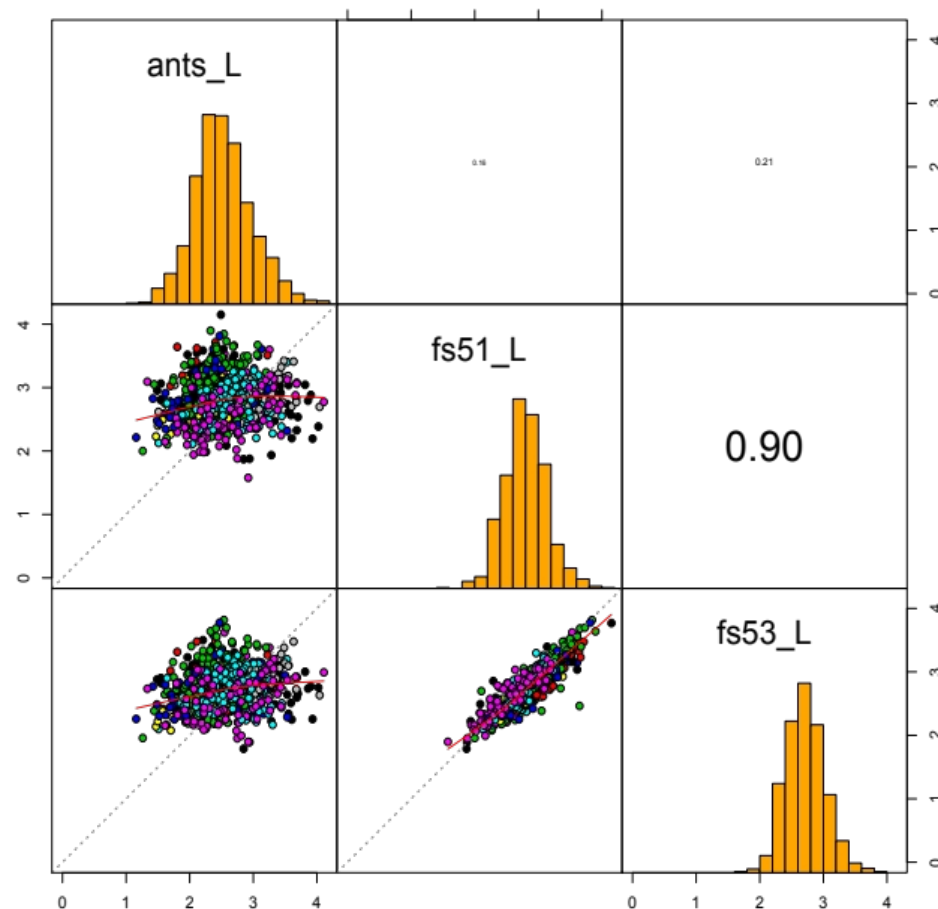
Across parametrizations of software and across software



Eklund et al., PNAS, 2016 :

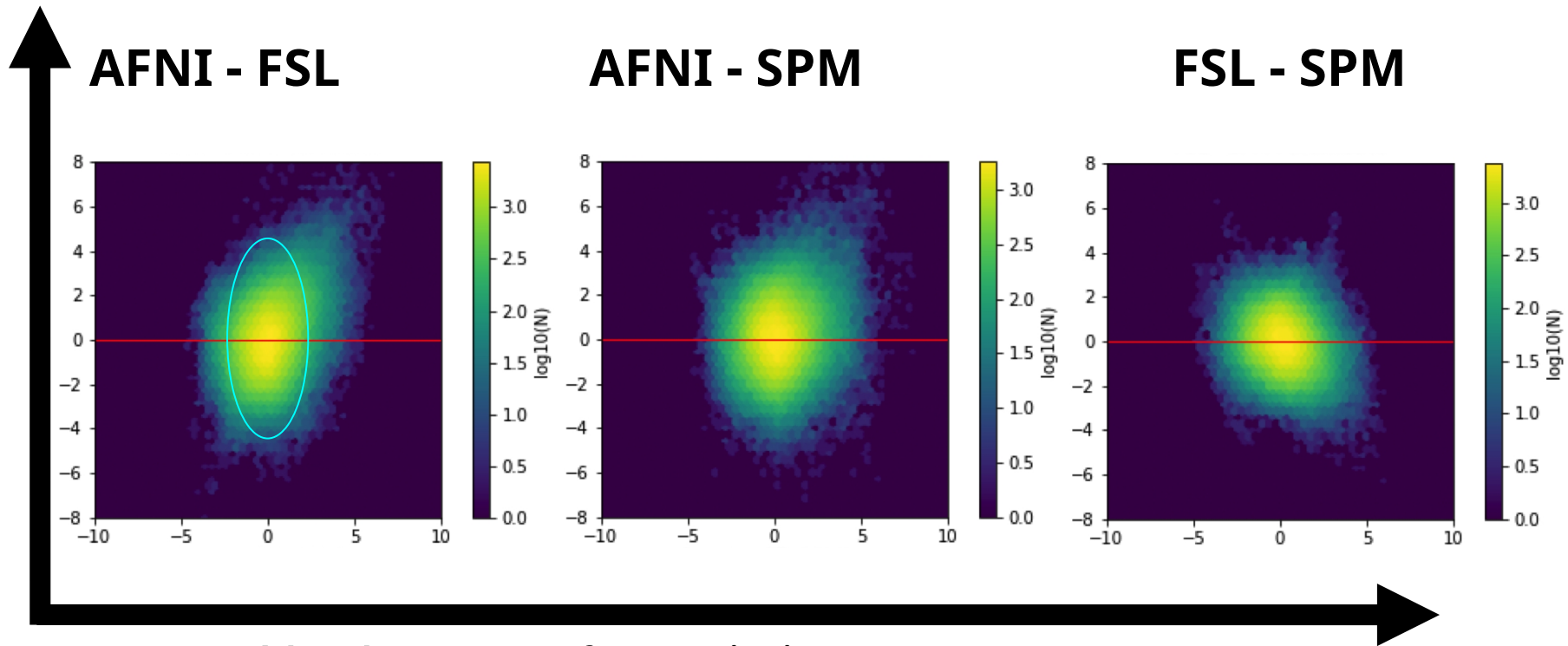
- Low threshold issue
- High threshold issue with Paradigm E1 ?
- Ad hoc procedure leads to around 70% FPR

- Estimated 3,500 papers affected by low threshold ?
- But 13000 w/o multiple comparisons ?

Size of the left caudal anterior
Cingulate

Dickie E et al., 2017

Y = Diff. of t-statistics

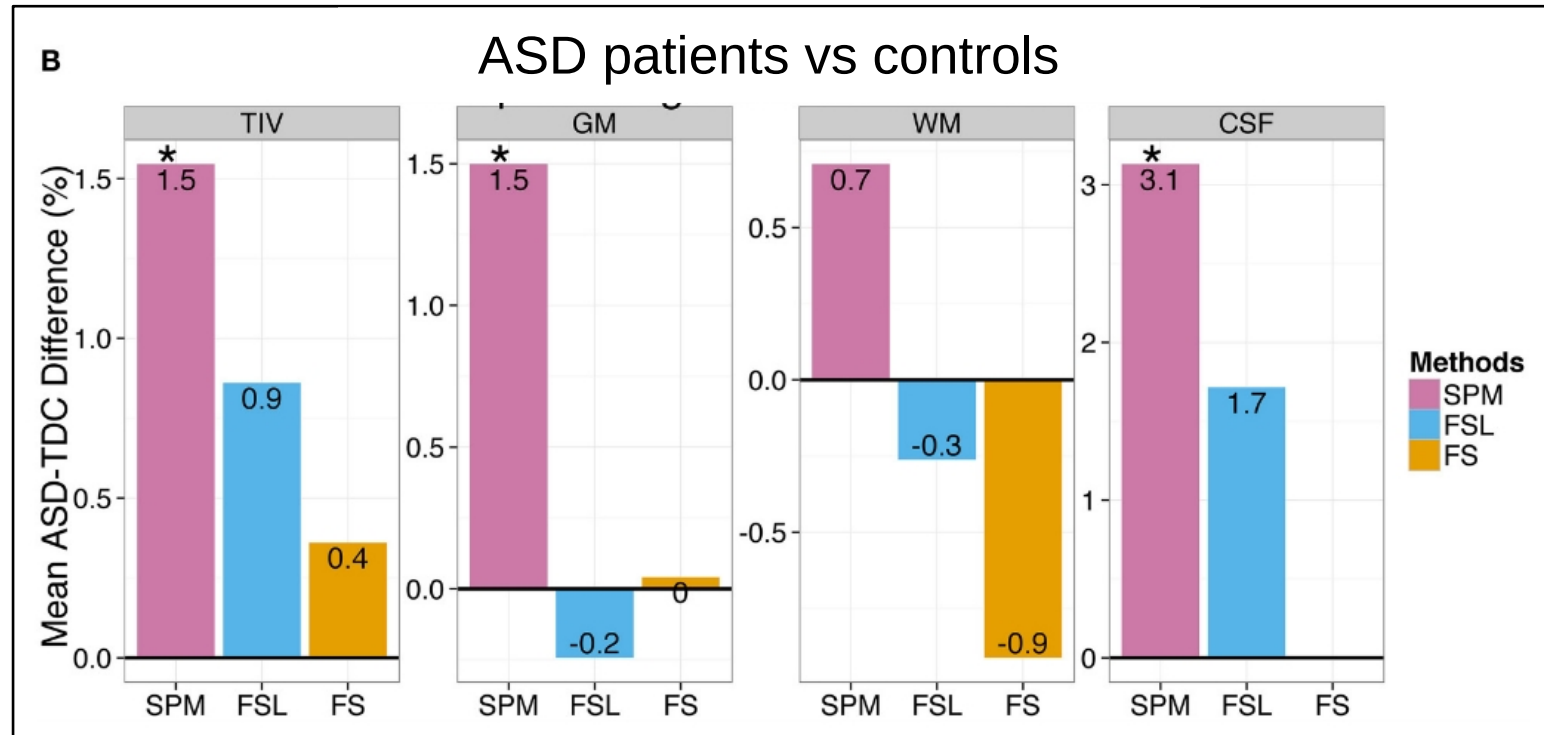


X = Average of t-statistics

- ▷ Plots similar to expected variation if **independent** was fed into each package

Alex Bowring, Camille Maumet, Thomas Nichols

G. Katuwal, f. in Brain Imaging Methods, 2016

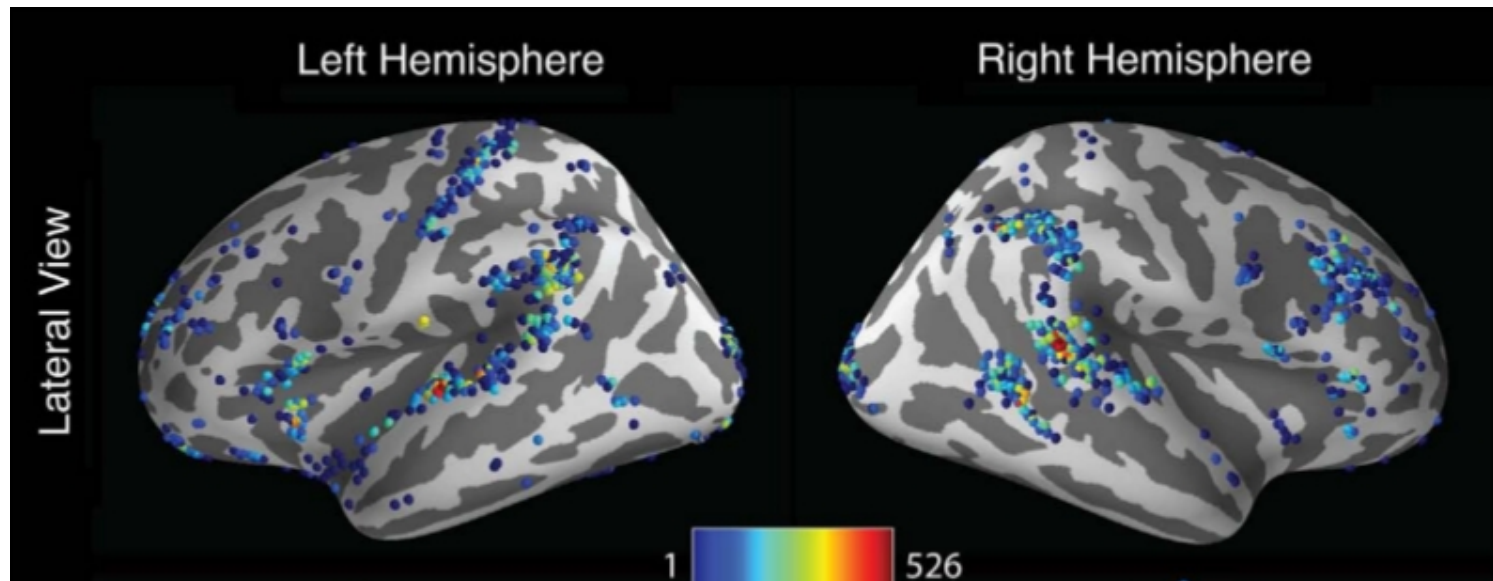


- Change from FSL to SPM?
- Change from v.1.12 to v.2.1 ?
- Change from cluster A to cluster B? Glatard et. al., finisc, 2015

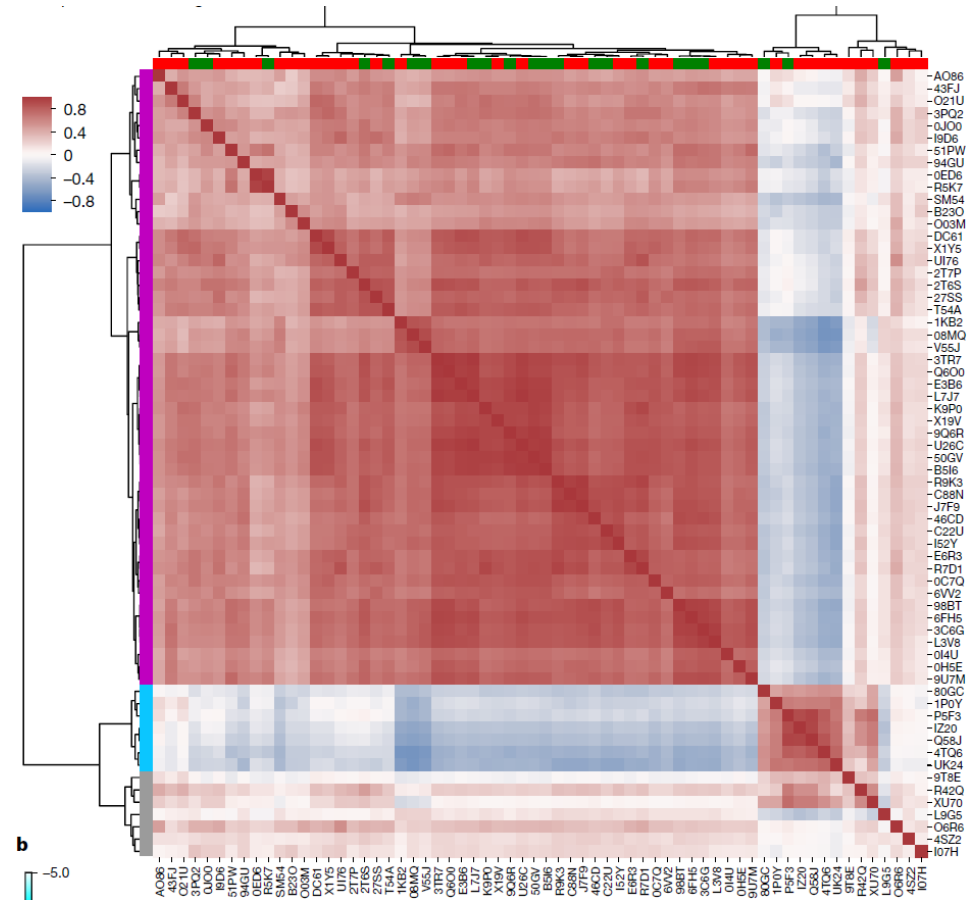
Across possible pipelines

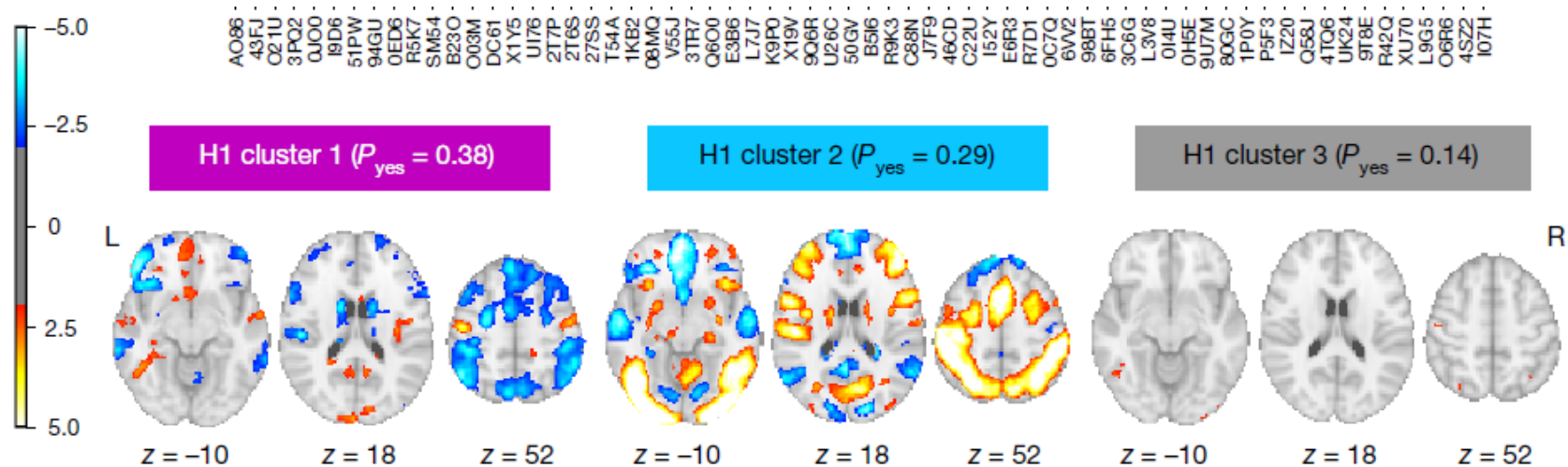
J. Carp, f. Neuroscience, 2012

- A **single** event-related fMRI experiment to a large number of unique analysis procedures
- Ten analysis steps for which multiple strategies appear in the literature : **6,912 pipelines**
- Plotting the maximum peak

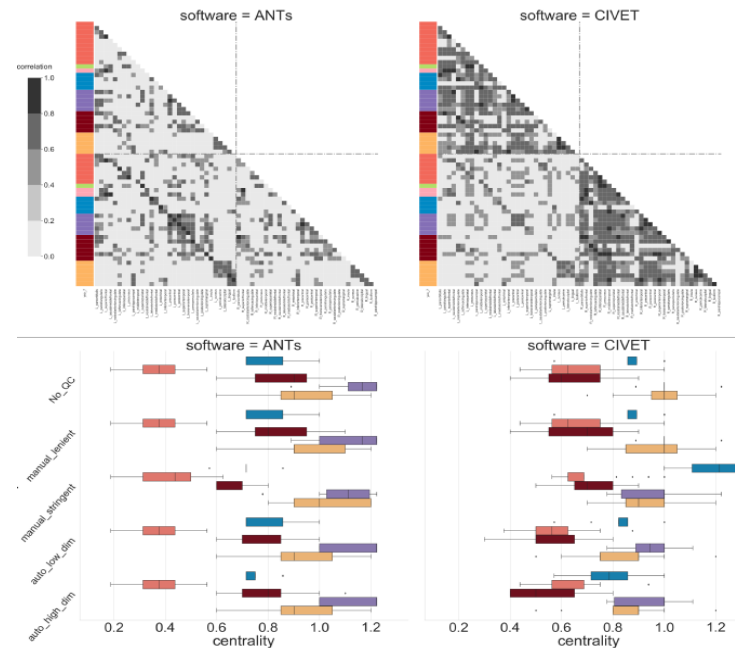


- 70 independent teams analyzing the same fMRI dataset
- No team had the same pipeline
- Results show three “clusters”
- Even within clusters decision to reject H_0 varies



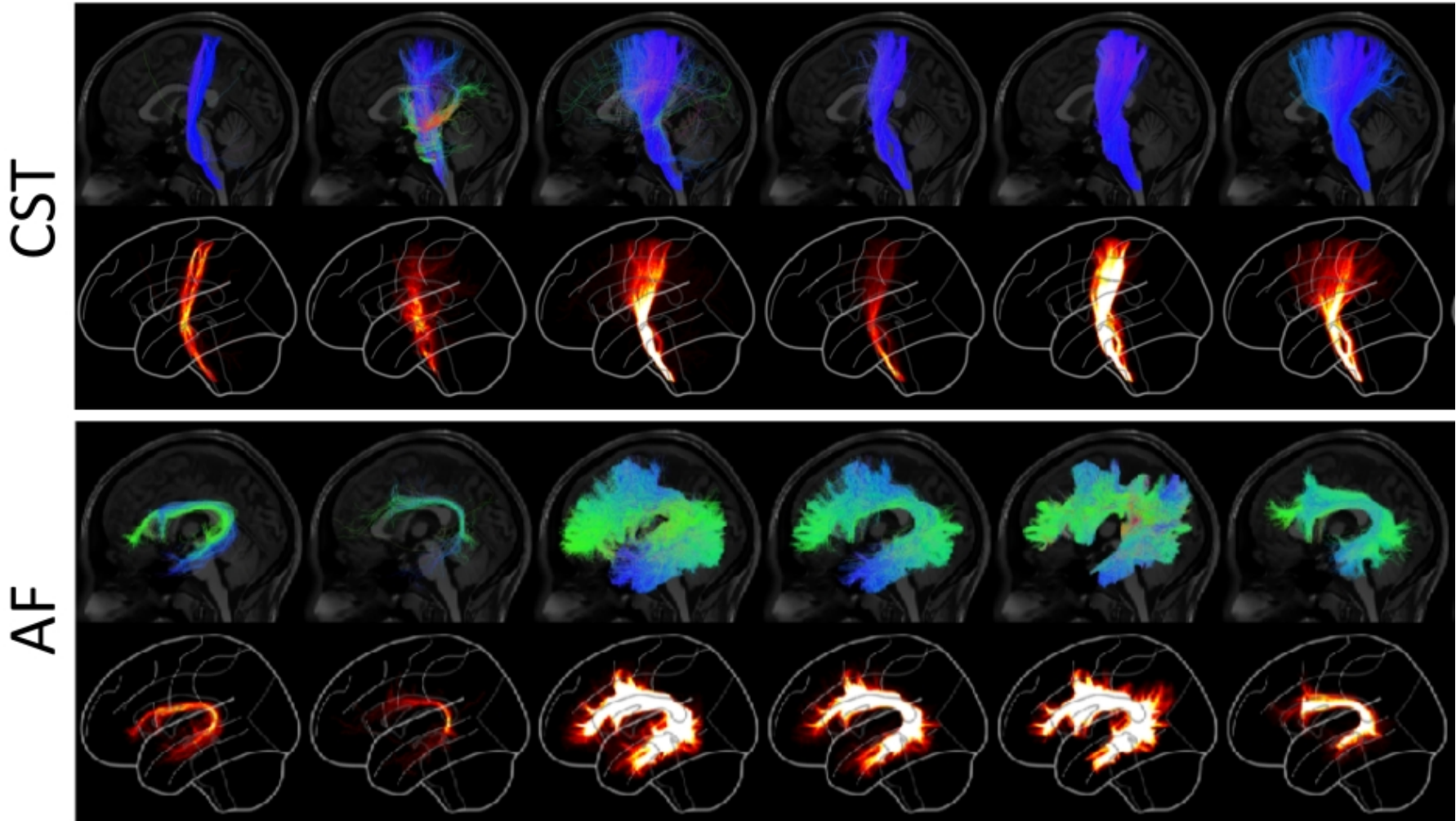


Analytical flexibility: N. Bhagwat, 2020



What happens when 42 groups dissect 14 white matter bundles on the same dataset?

KG Schilling ... M Descoteaux 2020 (~150 authors)



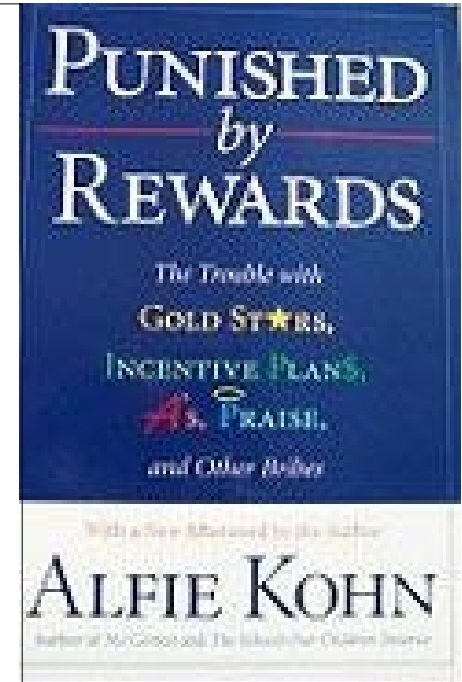
- A less rare case than usually thought !
- No license
- Database not containing what it describes
- Wrong QC – QC unreliable
- Headers of files are not correct (cf the Left/Right issue)
- Provenance of data is lost
- **SAM1 SAM2 SAM3:**
<https://www.youtube.com/watch?v=N2zK3sAtr-4>

- From HCP:

"With the releases of FreeSurfer 7.X, there have been some regressions in surface placement performance when running FreeSurfer inside the HCP Pipelines. At this time, *I would recommend sticking with FreeSurfer 6.0* while we get these issues sorted out."

1. Poor statistical procedures
2. Issues in data and software
3. A cultural issue: Publication practices and research incentives

- Publication = the only “currency” for researchers, universities
- The high competition incites researchers to keep data and code as “assets” and to get as many authorships as possible
- The current incentive system promotes poorly reproducible research



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Research



The natural selection of bad science

Paul E. Smaldino¹ and Richard McElreath²

¹Cognitive and Information Sciences, University of California, Merced, CA 95343, USA

²Department of Human Behavior, Ecology, and Culture, Max Planck Institute for Evolutionary Anthropology, Leipzig, Germany

Objectives US national pilot study to

- (1) test the feasibility of online administration of the Bioethical Issues in Biostatistical Consulting (BIBC) Questionnaire
- (2) determine the prevalence and relative severity of a broad array of bioethical violations requests that are presented to biostatisticians by investigators seeking biostatistical consultations; and
- (3) establish the sample size needed for a full-size phase II study.

Conclusion: **clear evidence** that researchers make requests of their biostatistical consultants that are rated as **severe violations** and occur **frequently**

Wang et al. 2017. BMJ Open 7 (11): 2017.

Publication
bias

Publish or
conduct
next
experiment

Generate
and specify
hypotheses

Failure to
control for
bias

Design
study

Low
statistical
power

Poor quality
control, e.g.
misidentified
cell lines/
reagents

Interpret
data

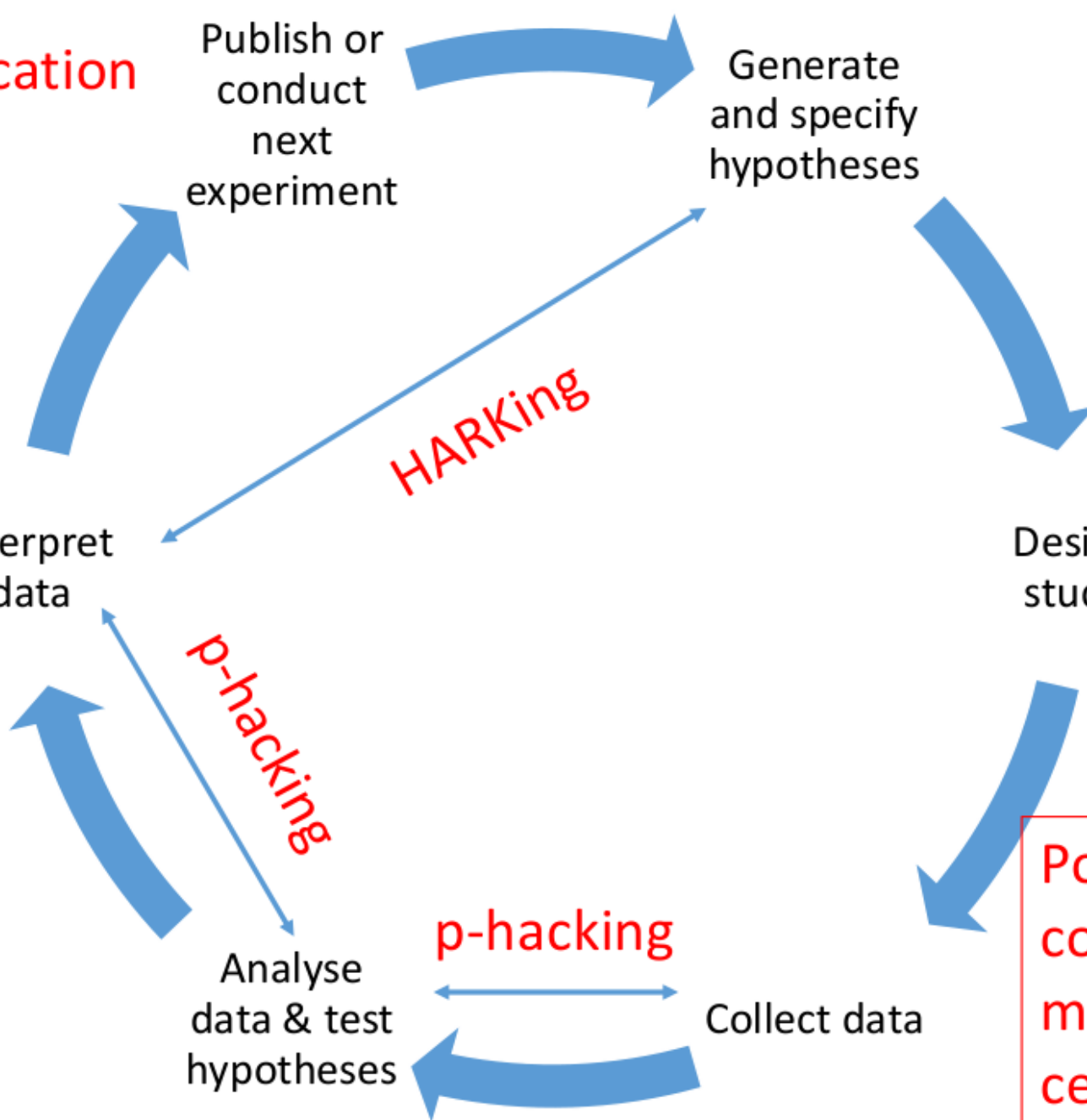
Analyse
data & test
hypotheses

Collect data

HARKing

p-hacking

p-hacking



Part I: Reproducibility: background

Part II : Etiology of Irreproducibility

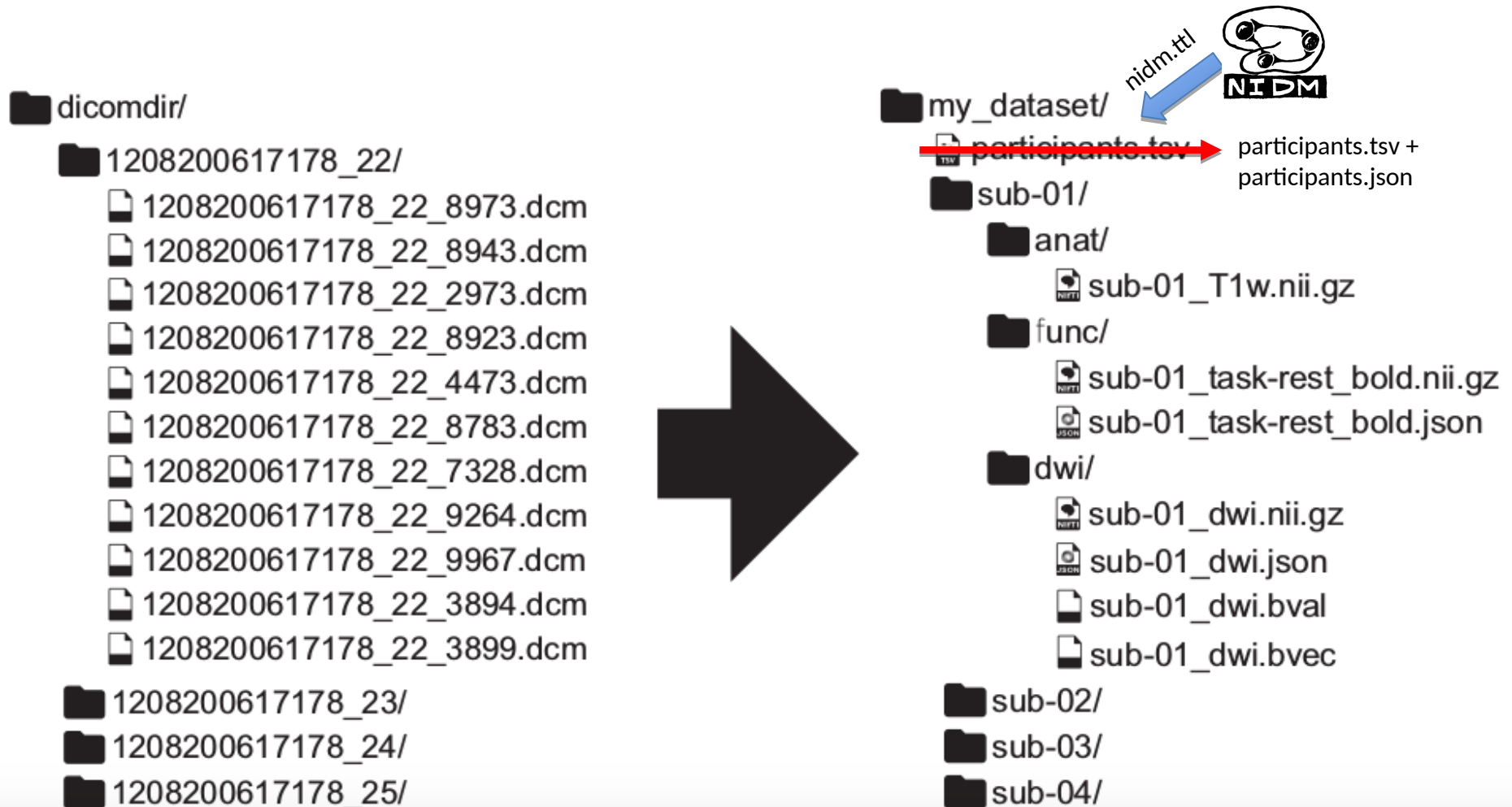
Part III : Some therapeutic proposals

- Make/use the tools that change the culture ex:
used standard to harmonize data
- Improve training and open practices
- Develop community, training & standards
- Change the incentives

schema.org

What if we had google power for datasets ?

BIDS / RDF



Participants.tsv file:

ADHD200 - Brown

sub-0026001

sub-0026002

sub-0026004

sub-0026005

Participants.tsv file:

participant_id	gender	age	handedness	verbal_iq	performance_iq	full4_iq	qc_rest_1	qc_anatomical_1
26001	Male	16.92	Right	133	104	120	Pass	Pass
26002	Male	15.68	Right	106	106	107	Pass	Pass
26004	Female	14.99	Right	119	123	125	Pass	Pass
26005	Female	15.16	Right	116	131	126	Pass	Pass
26009	Male	16.91	Left	113	81	97	Pass	Pass
26014	Female	16.21	Right	101	102	102	Pass	Pass
26015	Female	15.2	Right	127	98	113	Pass	Pass
26016	Male	16.07	Right	120	96	109	Pass	Pass
26017	Female	14.56	Right	95	87	89	Pass	Pass
26022	Male	17.83	Right	105	111	109	Pass	Pass
26024	Female	17.77	Right	89	83	85	Pass	Pass
26027	Female	11.28	Right	108	103	106	Pass	Pass
26030	Female	14.51	Right	121	119	123	Pass	Pass
26039	Female	14.19	Right	125	117	124	Pass	Pass
26040	Female	13.67	Right	111	93	102	Pass	Pass
26041	Female	13.68	Right	129	120	128	Pass	Questionable
26042	Female	12.82	Right	106	110	109	Pass	Pass

Describe data with standard terminology

Link data

RDF triples form (linked) graphs

<Bob> <is a> <person>.

<Bob> <is a friend of> <Alice>.

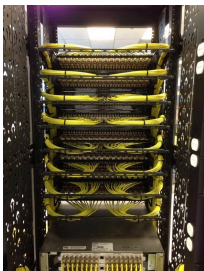
<Bob> <is born on> <the 4th of July 1990>.

<Bob> <is interested in> <the Mona Lisa>.

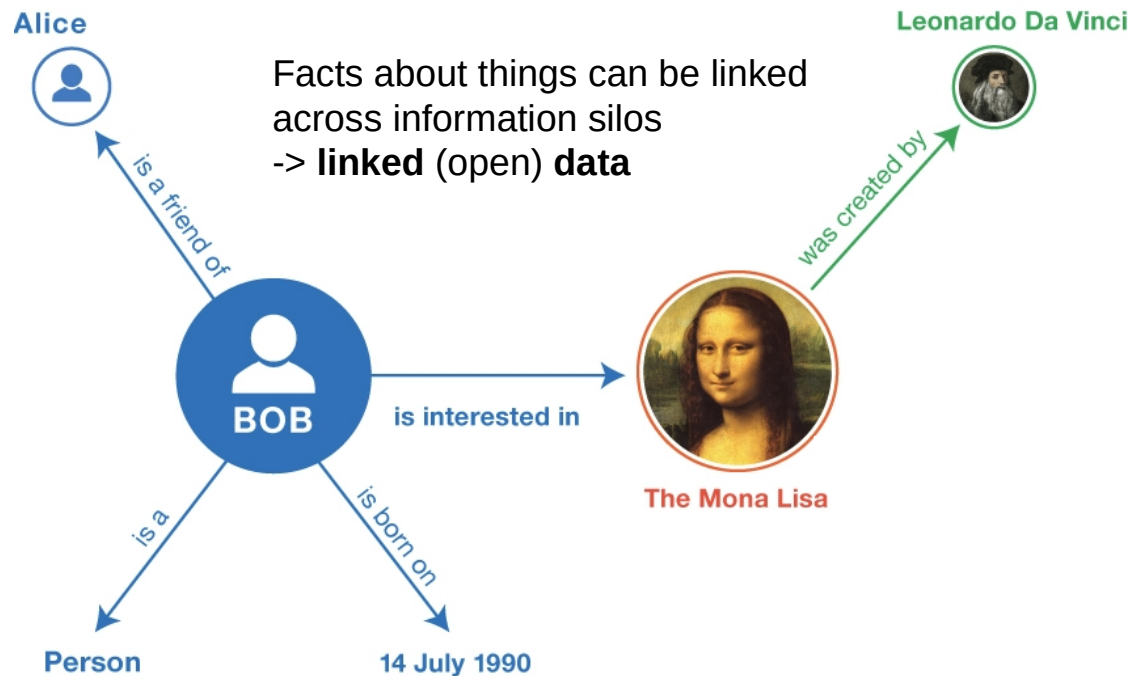
<the Mona Lisa> <was created by> <Leonardo da Vinci>.

vocabularies? -> many!

Theory



Practice



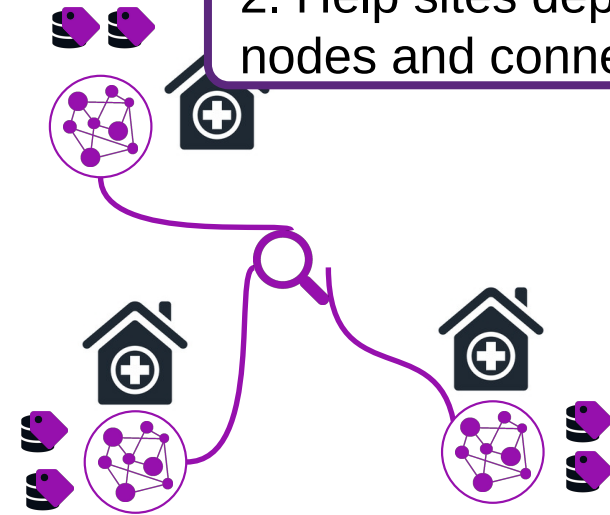
Global data findability and sharing

Neurobagel: query.neurobagel.org

1. Annotation with common data model

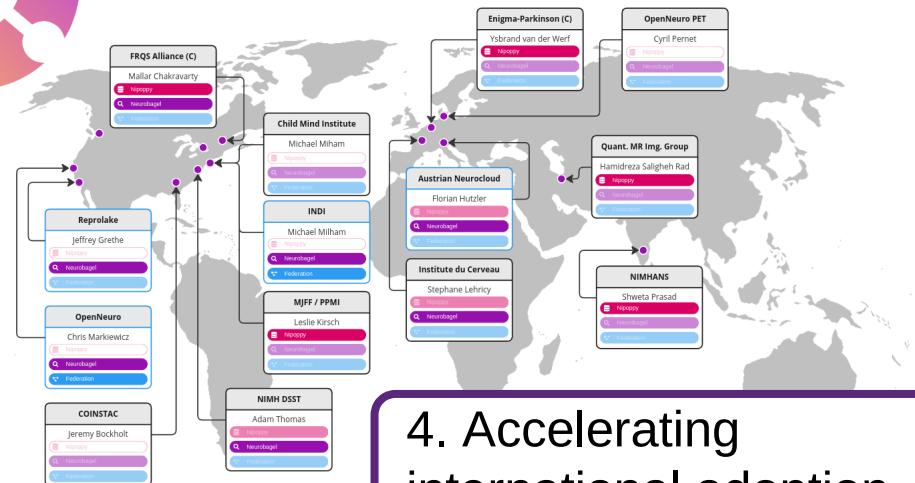
The screenshot shows the 'Neurobagel Annotation' interface. On the left, there's a sidebar with 'Annotated' and 'Unannotated' sections. The main area has a table with columns: 'Value', 'Description', 'Standardized Term', and 'Treat as missing value'. The 'Standardized Term' dropdown is set to 'autism'. The 'Treat as missing value' column has 'No' and 'Yes' buttons. At the bottom, there are 'Back to Column Annotation' and 'Next: Download' buttons.

2. Help sites deploy nodes and connect



3. Federated cohort discovery

The screenshot shows the 'Neurobagel Federated cohort discovery' interface. It features a search bar at the top with the text 'Define and find cohorts'. Below the search bar, there's a list of search results. The first result is 'Cognitive Control Theoretic Mechanisms of Real-time fMRI-Guided Neuromodulation (CTM)' from OpenNeuro, with 77 subjects match / 77 total subjects. The second result is 'Cognitive Training' from OpenNeuro, with 166 subjects match / 166 total subjects. The interface includes filters for 'Minimum number of imaging sessions', 'Minimum number of phenotypic sessions', 'Assessment tool', 'Imaging modality', 'Pipeline name', and 'Pipeline version'. At the bottom, there are 'Submit Query' and 'How to access data' buttons.



4. Accelerating international adoption

Tools / infrastructure

- How much should I invest in learning the tools ?
 - Which tools ? Can you take them with you?
 - What are your personal and technical constraints / priorities?
- Be part of a community with similar mindsets
 - You may be seating in an environment that is not conductive of reproducible research
- Re-think reproducibility in terms of efficiency for your PI next student if not for yourself

Tools / infrastructure

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What can we do ?

- Make the tools that change the culture
- **Improve open & reproducible practices**
- Develop community, training & standards
- Change the incentives

Pre-registration -registered report

- Promoted by many (eg Chris Chambers)
- Explain in details your hypothesis methods and software
- The pre-registered report is peer reviewed by a journal
- If accepted, results (even null results) will be published
- Does not preclude exploratory analyses
- A good idea even just for yourself

Is there any reason not to do it ?

Solution: Preregistration with OSF

There's a better way to manage your research

OSF is a free, open platform to support
your research and enable collaboration.

COMMENT • 10 SEPTEMBER 2019

What's next for Registered Reports?

Reviewing and accepting study plans before results are known can counter perverse incentives. Chris Chambers sets out three ways to improve the approach.

Chris Chambers 

- The rise of Git (GitHub, GitLab, ...) repositories
 - Many are open, collaborative
 - From lab-based “one software” to collaborative open source
- The rise of “FAIR” data format and containers
 - BIDS
 - BIDSapps
- The (slower) rise of standardized provenance information
 - NIDM - W3C prov



- Sustainability is in all the minds – governance?
- Critically linked to sustainability
 - 8. Be transparent in governance and operations
 - 9. Involve community in governance and decision making
 - 10. Be transparent on sustainability - financial and technical

OPEN
COMMENT

Recommendations for repositories and scientific gateways from a neuroscience perspective

It's the governance, stupid!: TAPIC: a governance framework to strengthen decision making and implementation [Internet]

Scott L. Greer¹, Nikolai Vasev², Holly Jarman¹, Matthias Wismar³, Josep Figueras³

Copenhagen (Denmark): European Observatory on Health Systems and Policies; 2019.
[European Observatory Policy Briefs.](#)

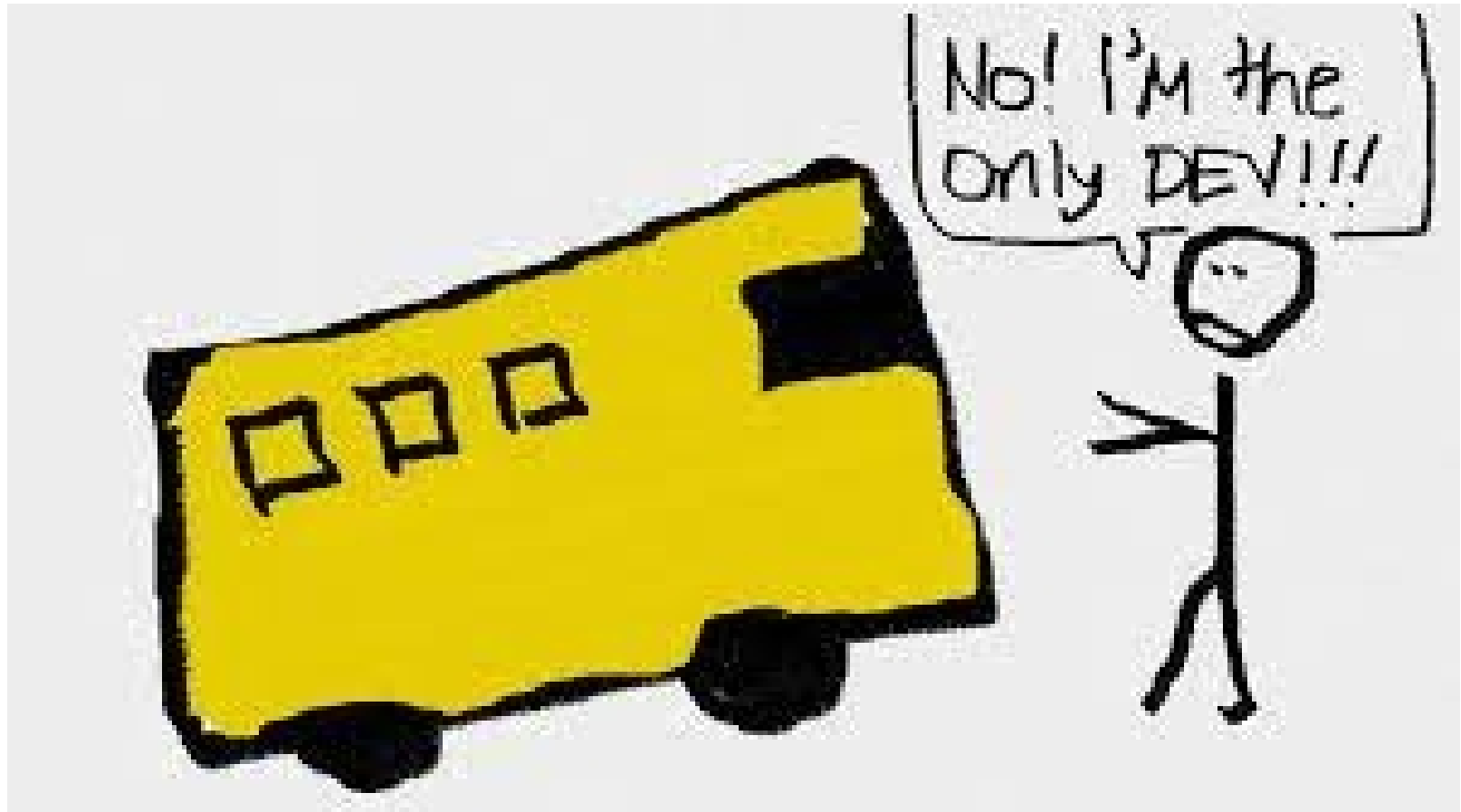
BRIDGE
Brain Research International Data Governance & Exchange

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Responsible and sustainable
governance frameworks for
for sharing brain data



Distributed development and governance: Nilearn



- Software / Code should receive equivalent level of scrutiny than narrative
- Software / Code should have an equivalent level of recognition
- Software need tests
 - Unit testing for refactorization
 - Simulated / known results for integration
- Infrastructures for reviewing and publishing notebooks

Building Research Platforms

SCIENTIFIC DATA

OPEN **Comment: High-quality science requires high-quality open data infrastructure**

Susanna-Assunta Sansone¹, Patricia Cruse² & Mark Thorley³

Resources for data management, discovery and (re)use are numerous and diverse, and more specifically we need data resources that enable the FAIR principles¹ of Findability, Accessibility, Interoperability and Reusability of data.

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Accepted: 29 January 2018

Published: 27 February 2018



OPEN **Recommendations for repositories and scientific gateways from a neuroscience perspective**

COMMENT

Malin Sandström¹  , Mathew Abrams¹, Jan G. Bjaalie² , Mona Hicks³, David N. Kennedy⁴ , Arvind Kumar⁵ , Jean-Baptiste Poline⁶ , Prasun K. Roy⁷, Paul Tiesinga⁸, Thomas Wachtler⁹  & Wojtek J. Goscinski^{10,11}

- Make the tools that change the culture
- Improve training and open practices
- **Develop community, training & standards**
- Change the incentives

- The rise of Git (GitHub, GitLab, ...) repositories
 - Many are open, collaborative
 - From lab-based “one software” to collaborative open source
- The rise of “FAIR” containers
 - BIDS
- The (slow) rise of standardized provenance information
 - NIDM - W3C prov

Training: NIH P41 ReproNim

ReproNim

HOME

PROJECTS

TRAINING

CONTACT

*ReproNim: A Center for Reproducible
Neuroimaging Computation*

(Discover, Replicate, Innovate)^{Repeat}

TELL ME MORE

REPRONIM INTRO

REPRODUCIBILITY BASICS

FAIR DATA

DATA PROCESSING

STATISTICS

Data Science

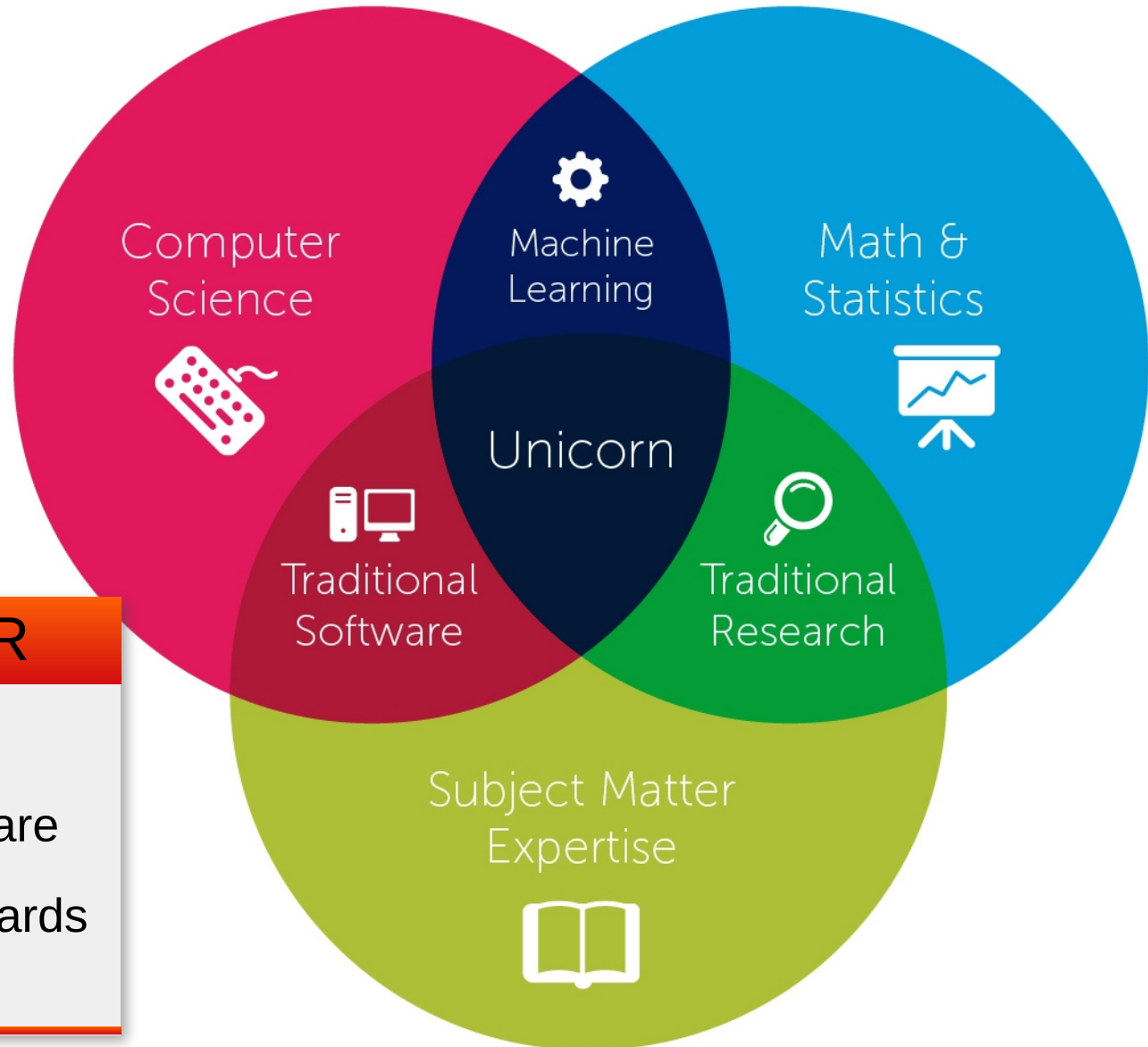
Ethical Scholarly communications
Epistemology / lessons from the past
How to collaborate and teach

FAIR

Data

Software

Standards
reuse



Science and Technology

Research

- Make the tools that change the culture
- Improve training and open practices
- Develop community standards
- Change the incentives

Changing the publication model



Reproducibility: A tragedy of errors, Allison et al, 2016, Nature

Data & Software research objects

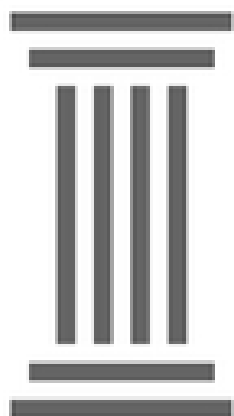
DATA	SOFTWARE
DataCite	CiteSoftware
FORCE11 JDDCP	software.ADORE
Scientific Data / ...	JOSS/ ...
Data packages	Containers
Governance (GDPR vs OpenNeuro?)	Governance (Scipy vs Numpy ?)

What is needed ?

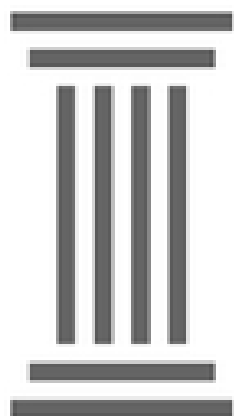


**An open publishing platform for
the neuroscience community**

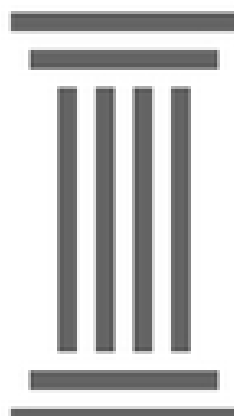
APERTURE



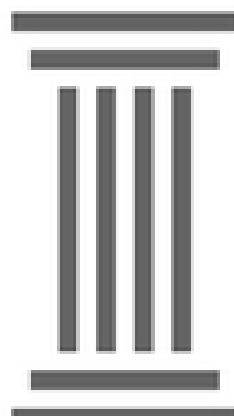
**High-Quality
Publications**



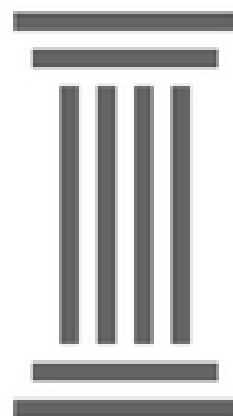
Open-Access



Low Cost



**Diversity of
published
Research Objects**



**Community
Driven**

GENERATIVE AI

Come try Copilot!

Monday, May 25, 1:00 - 2:00pm (online)

Curious about integrating AI into your teaching or exploring its possibilities for learning? Join us for this beginner-friendly session, to try out Copilot, McGill's approved generative AI tool. [Register now!](#)

Better prompts with Copilot

Wednesday, May 27, 1:00 - 2:00pm (online)

This guided session dives deeper into crafting more effective and intentional prompts. Work through examples and teaching-related scenarios to explore advanced prompting techniques. [Register now!](#)

Sharing experiences with generative AI: Teaching and Learning Community

Thursday, May 28, 1:00 - 2:00pm (in person)

This Community provides a space to exchange ideas and build connections with colleagues exploring similar questions around the integration of AI in higher education.

[Learn more and register.](#)

(Re)Designing assessments in the age of gen AI

Friday, May 29, 10:00am - 12:00pm (online)

Whether the goal is to integrate AI into their assessments, or to design assessments that minimize its use, this session will provide frameworks, examples, and collaborative discussion.

[Register now!](#)

Thank you

- Lab@McGill: <https://neurodatascience.github.io/>
- **McGill-Concordia-UdM** colleagues: T. Glatard, P. Bellec, C. Greenwood, A. DeGuise and many others
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- **Berkeley** colleagues: S. Van der Walt, M. Brett, J. Millman, Dan Lurie, M. D'Esposito, et al
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